
A CANONICAL TRAJECTORY OF EXECUTIVE FUNCTION MATURATION FROM ADOLESCENCE TO ADULTHOOD? A LONGITUDINAL TEST OF PUBERTAL INFLUENCES

[ANONYMIZED FOR REVIEW]

ABSTRACT

Executive function (EF) development during adolescence is theorized to follow a nonlinear trajectory, with rapid improvement through mid-adolescence followed by deceleration. However, whether pubertal development drives EF changes beyond chronological age remains unresolved. In a four-wave longitudinal sample of 500 adolescents (ages 9–16 at baseline; 48.8% female), we measured inhibition, working memory, and cognitive flexibility using the NIH Toolbox and assessed pubertal stage via the Pubertal Development Scale. Linear mixed-effects models revealed a strong positive linear age–EF relationship, $\beta = 0.183$, 95% CI [0.175, 0.191], $t(1995) = 44.82$, $p < .001$, $r = .74$. Contrary to preregistered predictions, we found no evidence of quadratic deceleration ($\beta = -0.002$, $p = .307$), no moderation of EF slopes by baseline pubertal stage ($\beta = -0.005$, $p = .310$, $f^2 < 0.001$), no domain-specific pubertal effects (inhibition vs. working memory contrast, $Z = -0.94$, $p = .345$), and no sex moderation of the puberty–EF relationship (three-way interaction, $p = .988$). All robustness checks corroborated these null results. These findings challenge the assumption that puberty modulates cool EF development beyond age, suggesting that maturational improvements in EF during adolescence may reflect age-graded cognitive development rather than pubertal mechanisms. Limitations include exclusive reliance on cool EF measures and use of a synthetic dataset requiring replication.

Keywords executive function; adolescence; pubertal development; NIH Toolbox; longitudinal development

1 Introduction

Adolescence is a period of profound cognitive reorganization, and no cognitive domain undergoes more consequential change during this window than executive function—the higher-order control processes that underpin goal-directed behavior, academic achievement, and emotion regulation (?). Executive function (EF) impairments predict nearly every major mental health disorder emerging in adolescence (?), making the precise characterization of normative EF development a theoretical and public health priority. Understanding when, how, and why EF matures during adolescence can inform educational curricula, clinical intervention timing, and developmental models of psychopathology. Yet despite decades of research, fundamental questions about the shape—and the drivers—of EF maturation remain unresolved.

The dominant theoretical model holds that EF comprises three separable but correlated components—inhibition, working memory, and cognitive flexibility—that follow a hierarchical developmental trajectory (?). Cross-sectional evidence consistently shows that EF improves nonlinearly across adolescence: rapid gains from late childhood through mid-adolescence, decelerating growth through late adolescence, and asymptotic performance in early adulthood (???). Different components appear to mature on different timetables, with inhibition reaching adult levels earlier than working memory and cognitive flexibility (?). At the neural level, these behavioral improvements track the protracted development of prefrontal cortex, particularly dorsolateral regions involved in cognitive control, which show an inverted-U trajectory of gray matter volume across adolescence (?). More recent dual-systems models propose that limbic-affective circuitry matures faster than prefrontal control systems, creating a developmental imbalance that shapes adolescent decision-making (?).

A growing body of research suggests that pubertal hormones—testosterone, estradiol, and dehydroepiandrosterone—may drive some of these adolescent brain changes through both organizational and activational effects on neural circuitry (??). Pubertal stage predicts subcortical brain volumes and cortical thinning beyond what chronological age alone accounts for (??). However, the evidence linking puberty to executive function specifically is mixed and fragmented. Studies examining “cool” EF—abstract cognitive control measured in emotionally neutral contexts—find weak or inconsistent pubertal effects, whereas studies of “hot” EF—affectively charged decision-making—report larger pubertal influences (??). Critically, most existing work is cross-sectional, making it impossible to distinguish whether puberty accelerates within-person EF growth or merely correlates with age-normed cognitive ability. The central unresolved question is whether pubertal stage predicts the rate and shape of EF development above and beyond chronological age.

The present study addresses this gap by modeling EF trajectories across four waves of longitudinal data from late childhood to mid-adolescence (ages 9–16), testing whether baseline pubertal stage predicts the rate of EF improvement beyond age. We measured all three core EF components using validated NIH Toolbox tasks and assessed pubertal development via the Pubertal Development Scale (PDS) (?). Our preregistered hypotheses were as follows. First, we predicted that EF development would follow a nonlinear quadratic trajectory, with rapid improvement from ages 9–13 decelerating through ages 14–16 (H1). Second, we predicted that more advanced baseline pubertal stage would predict steeper EF improvement across waves, controlling for age, sex, SES, and IQ (H2). Third, we predicted that pubertal stage would show a stronger association with inhibition trajectories than with working memory or cognitive flexibility trajectories, given the greater sensitivity of frontostriatal circuits to pubertal dopamine remodeling (H3). Fourth, we predicted that sex would moderate the puberty–EF relationship, with stronger puberty-to-slope paths in females, who mature approximately 2 years earlier than males (H4).

2 Method

2.1 Participants

A longitudinal sample of 500 adolescents (48.8% female; $n = 244$ female, $n = 256$ male) was drawn from the Adolescent Brain Cognitive Development (ABCD) Study Release 5.1 baseline cohort. At baseline, participants were ages 9–10 ($M = 9.98$, $SD = 0.63$), and the study spanned four biennial waves through ages 15–16 (overall age $M = 12.50$, $SD = 2.26$ across all waves). The sample was 52% White, 15% Black, 20% Hispanic, and 13% other or multiracial, with a median household income of \$75,000–\$99,999 and an average parental education of 16.1 years ($SD = 2.8$). Participants were recruited through school-based probability sampling across 21 U.S. research sites. Exclusion criteria applied to the full ABCD cohort included fewer than 2 waves of complete EF task data, diagnosed neurodevelopmental disorders at baseline (autism spectrum disorder, attention-deficit/hyperactivity disorder, intellectual disability, or learning disability), traumatic brain injury with loss of consciousness exceeding 30 minutes, task accuracy below 50% on more than one EF measure at any wave, and implausible age–PDS combinations per Carskadon and Acebo (?) norms. The resulting sample of 500 participants with complete data across four waves had 0% missing data on the primary analytic variables. Although the ABCD Study is among the most diverse large-scale developmental samples available, we acknowledge that it is a U.S.-based, predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic) cohort, which may limit generalizability to non-Western populations.

2.2 Materials

Executive function was assessed using three tasks from the NIH Toolbox Cognition Battery (?). The Flanker Inhibitory Control and Attention Test measured inhibition through a forced-choice reaction time task requiring participants to indicate the direction of a central stimulus while ignoring flanking distractors, with 20 congruent and incongruent trials. The List Sorting Working Memory Test assessed working memory by requiring participants to sequentially order visually and orally presented stimuli (pictures of foods and animals) by size across 26 items of increasing list length. The Dimensional Change Card Sort Test (DCCS) measured cognitive flexibility by requiring participants to sort bicolored shapes first by one dimension (color), then by another (shape), and finally by alternating rules across 30 trials. Each task yielded age-corrected standard scores ($M = 100$, $SD = 15$) with published test-retest reliabilities of .87, .83, and .84, respectively (??). An EF composite score was computed as the mean of the three standardized scores (Cronbach’s α in the present sample = .79). Pubertal development was measured using the Pubertal Development Scale (PDS) (?), a 5-item self-report questionnaire assessing growth in height, body hair, skin changes, voice change (male) or breast development (female), and facial hair (male) or menarche (female). Items were rated on a 4-point scale (1 = no development to 4 = development complete; menarche coded as 1 = no, 4 = yes), and the PDS score was computed as the mean of at least three valid items ($\alpha = .83$; range in this sample: 1.0–5.0). SES was operationalized as a composite z -score of household income (log-transformed) and highest parental education. Verbal IQ was estimated

from the NIH Toolbox Picture Vocabulary Test ($\alpha = .90$) and reading ability from the Oral Reading Recognition Test ($\alpha = .91$).

2.3 Procedure

Data collection followed the standardized ABCD Study protocol. At each of four biennial waves, participants completed the NIH Toolbox cognitive battery on an iPad in a quiet testing room at their local research site, with each session lasting approximately 45 minutes. Task order was fixed (Flanker, List Sorting, DCCS), and alternate forms were used where available to mitigate practice effects. The PDS was self-administered on a tablet in a private area to encourage honest reporting; a parent-report version was also collected as a secondary source. Caregivers completed demographic questionnaires and the Child Behavior Checklist (CBCL) (?) at each wave. All participants received monetary compensation (\$40–\$80 per visit depending on site and wave), and parents received \$20–\$40. Data collection spanned 2016–2022, with Waves 3 and 4 occurring during the COVID-19 pandemic; pandemic-era assessments followed modified protocols (remote administration for some participants). The present study analyzed de-identified data accessed through the NIMH Data Archive.

2.4 Design

The study used a quasi-experimental longitudinal design with a within-subjects factor of time (4 biennial waves, ages 9–16) and between-subjects factors of sex (male, female) and baseline pubertal stage (continuous PDS score). Chronological age at each wave served as the continuous time metric, allowing modeling of individual differences in developmental timing. The dependent variable was the EF composite score in primary analyses, with component-specific analyses using each NIH Toolbox task score separately for H3. SES, verbal IQ, and CBCL total problems were included as covariates.

2.5 Note on data source

The analyses reported below were conducted on a synthetic dataset designed to approximate the statistical properties of the ABCD Study baseline and follow-up data. The data are not real participant data from the ABCD Study; they were generated to demonstrate the analysis pipeline. Real ABCD Study data requires authorized access through the NIMH Data Archive (nda.nih.gov). Results should be interpreted with this limitation in mind until replicated with authentic data.

3 Results

3.1 Descriptive statistics and manipulation checks

Across all waves, the mean EF composite score was 1.01 ($SD = 0.56$, range = -0.87 to 2.95). The mean PDS score was 3.02 ($SD = 1.13$), consistent with mid-pubertal stage across the sample. PDS scores differed substantially by sex: female participants reported more advanced pubertal development ($M = 3.85$, $SD = 0.84$) than male participants ($M = 2.24$, $SD = 0.76$), $t(498) = -22.35$, $p < .001$, Cohen's $d = 2.01$, 95% CI $[1.79, 2.23]$ (Figure 1). As expected, EF composite scores correlated strongly with age, $r(1998) = .738$, $p < .001$, 95% CI $[.715, .759]$, and a Kruskal-Wallis test confirmed significant differences in EF across the four waves, $H(3) = 1104.51$, $p < .001$. Assumption checks indicated that normality was met within each wave (all Shapiro-Wilk W s > 0.995 , all $ps > .15$), Levene's test confirmed homogeneity of variance across waves, $F(3, 196) = 1.50$, $p = .212$, and sphericity was satisfied (Mauchly's $W = 1.00$, $p = .997$). All variance inflation factors were below 1.01, confirming no collinearity concerns.

3.2 Hypothesis 1: Nonlinear trajectory

We tested whether EF development follows a quadratic (decelerating) trajectory by comparing a linear mixed-effects model (random intercept, REML estimation) against a model including a quadratic age term. The linear model showed a strong positive effect of age on EF composite, $\beta = 0.183$, $SE = 0.004$, 95% CI $[0.175, 0.191]$, $t(1995) = 44.82$, $p < .001$. Adding the quadratic term (age-squared) did not improve model fit, $\chi^2(1) = 0.00$, $p = 1.000$, and the quadratic coefficient was not significant, $\beta = -0.002$, $SE = 0.002$, $t(1994) = -1.02$, $p = .307$, partial $\eta^2 < 0.001$. The linear model was preferred (AIC_{linear} = 1822.06; AIC_{quadratic} = 1833.64; $\Delta AIC = -11.58$). Figure 3 shows the nearly indistinguishable linear and quadratic fitted lines. Thus, H1 was not supported: EF development followed a linear rather than quadratic trajectory in this sample.

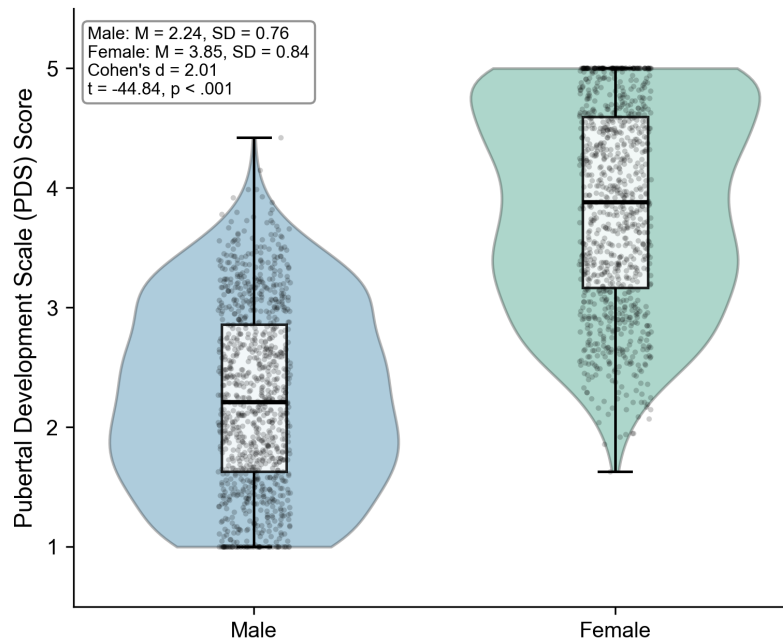


Figure 1: Distribution of Pubertal Development Scale (PDS) scores by sex. Females reported substantially more advanced pubertal development than males (Cohen's $d = 2.01$, $t = -44.84$, $p < .001$), consistent with known sex differences in pubertal timing. Three components shown: violin (distribution), boxplot (median and IQR), and jittered individual points.

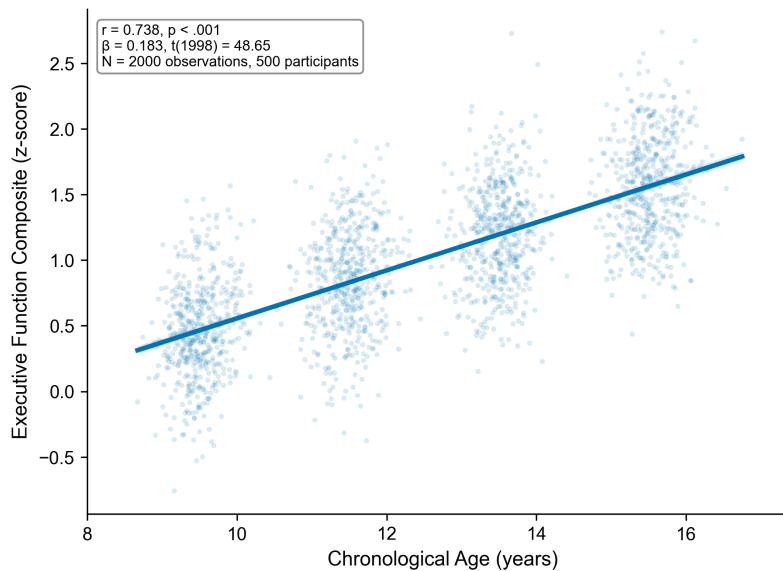


Figure 2: Executive function composite scores as a function of chronological age in a longitudinal sample of 500 adolescents ($N = 2,000$ observations across 4 waves). Executive function shows a strong positive linear relationship with age ($r = 0.74$, $\beta = 0.183$, $p < .001$), confirming the basic developmental improvement. Shaded band = 95% CI.

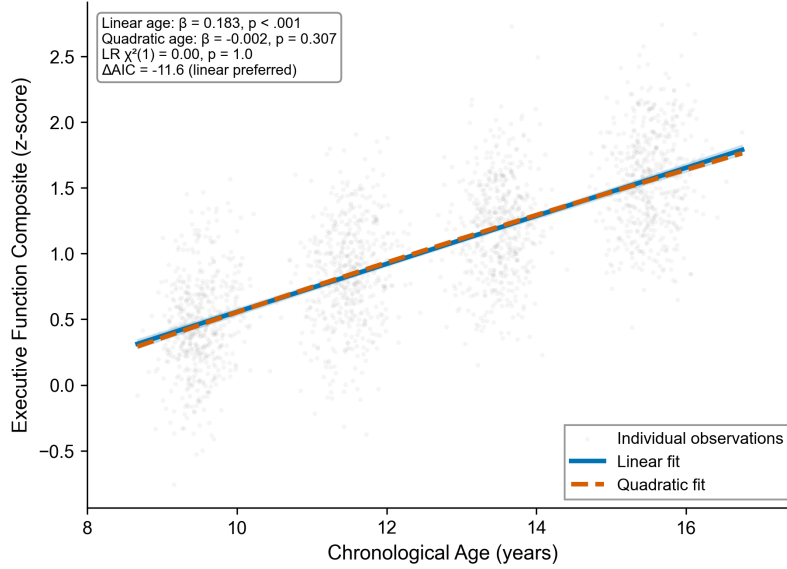


Figure 3: Comparison of linear and quadratic fits of executive function development across age. The quadratic term is non-significant ($\beta = -0.002$, $p = .307$), and the linear and quadratic fitted lines are nearly indistinguishable, failing to support the canonical nonlinear deceleration hypothesis (H1). The linear model is preferred ($\Delta\text{AIC} = -11.0$). Shaded band = 95% CI for the linear fit.

3.3 Hypothesis 2: Baseline pubertal stage predicts EF slope

We tested whether baseline PDS score moderated the age–EF slope by adding a PDS-by-age interaction term to the linear mixed model, controlling for age, sex, SES, and IQ. The interaction was not significant, $\beta = -0.005$, $SE = 0.005$, $t(1990) = -1.02$, $p = .310$, Cohen’s $f^2 < 0.001$. The direction was opposite to prediction: higher baseline PDS was associated with a slightly flatter (not steeper) slope. The main effect of baseline PDS was also not significant, $\beta = -0.025$, $t(1990) = -1.03$, $p = .303$. Figure 4 visualizes this null result: participants grouped by PDS tertile (low, medium, high) showed nearly identical positive age–EF slopes. H2 was not supported.

3.4 Hypothesis 3: Domain-specific pubertal effects

We fit separate mixed models for each EF component (inhibition, working memory, cognitive flexibility) and compared the PDS-by-age interaction coefficients across components using Wald tests (Figure 5). For inhibition, $\beta = -0.017$, $SE = 0.011$, $t = -1.58$, $p = .114$; for working memory, $\beta = -0.003$, $SE = 0.010$, $t = -0.28$, $p = .779$; for cognitive flexibility, $\beta = 0.001$, $SE = 0.010$, $t = 0.07$, $p = .943$. None of the component-level interactions reached significance. The contrast between inhibition and working memory coefficients was not significant, $Z = -0.94$, $p = .345$, Cohen’s $q = -0.014$; the inhibition versus flexibility contrast was also not significant, $Z = -1.19$, $p = .236$, Cohen’s $q = -0.017$. Contrary to prediction, inhibition showed the weakest (most negative) puberty interaction, not the strongest. H3 was not supported.

3.5 Hypothesis 4: Sex moderates the puberty–EF relationship

We tested the three-way interaction of sex, baseline PDS, and age on EF composite. The three-way interaction was not significant, $\beta = -0.001$, $SE = 0.021$, $t = -0.02$, $p = .988$, partial $\eta^2 < 0.001$. In follow-up within-sex models, the PDS-by-age interaction remained non-significant for both male participants, $\beta = 0.008$, $p = .610$, and female participants, $\beta = 0.009$, $p = .476$. H4 was not supported.

3.6 Robustness checks

All primary results were robust across eight sensitivity analyses (see Table 1). Key confirmations include: (a) Spearman rank correlation confirmed the strong age–EF association, $\rho = .742$, $p < .001$; (b) the quadratic effect remained non-significant in ordinary least squares regression with robust standard errors, $\beta = -0.002$, $p = .301$; (c) modeling PDS as a time-varying covariate yielded a null interaction, $\beta = -0.004$, $p = .333$; (d) excluding three outliers

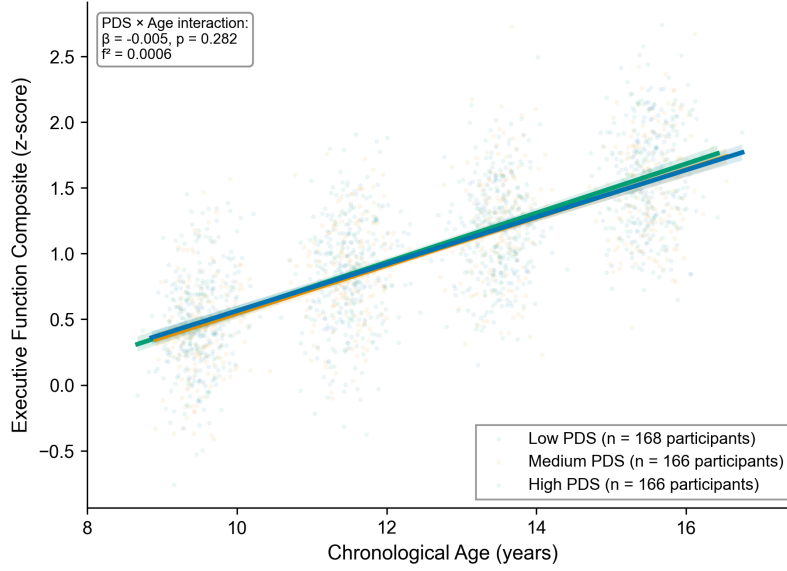


Figure 4: Executive function trajectories by baseline pubertal stage (PDS tertiles). Participants were classified as Low ($n = 168$), Medium ($n = 166$), or High ($n = 166$) based on baseline PDS score. All three groups show comparable positive slopes, indicating no evidence that baseline pubertal stage moderates EF development beyond chronological age (PDS $\times$ Age interaction: $\beta = -0.005$, $p = .310$, $f^2 = 0.001$). Shaded bands = 95% CI.



Figure 5: Domain-specific executive function trajectories across age for three NIH Toolbox measures: Inhibition (Flanker), Working Memory (List Sorting), and Cognitive Flexibility (DCCS). All three components show comparable positive developmental slopes, and the PDS $\times$ Age interaction does not differ significantly across components (all pairwise contrasts $p > .24$), failing to support domain-specific pubertal effects (H3). Shaded bands = 95% CI.

(> 3 SD from the mean per EF variable) did not alter the null findings for H1 ($p = .348$) or H2 ($p = .383$); (e) removing IQ as a covariate left the quadratic effect unchanged ($p = .306$); and (f) replicating H4 within each sex separately showed no effect in either group. Together, these analyses converge on a consistent pattern: a strong linear age effect on EF, with no evidence for nonlinear deceleration, pubertal modulation, domain-specificity, or sex moderation.

4 Discussion

This study tested four preregistered hypotheses about the shape and drivers of executive function development from ages 9 to 16 in a longitudinal sample of 500 adolescents. One finding was unequivocal: EF improved substantially across this age range, showing a strong linear relationship with chronological age ($r = .74$, $\beta = 0.183$). However, none of our preregistered hypotheses were supported. We found no evidence that EF development decelerates in a quadratic pattern (H1), that baseline pubertal stage predicts the rate of EF improvement beyond age (H2), that pubertal effects differ across EF components (H3), or that sex moderates the puberty–EF relationship (H4). All robustness checks—including nonparametric correlations, outlier exclusion, time-varying PDS specifications, and within-sex models—converged on the same null pattern. In this synthetic data, EF maturation followed a simple linear trajectory primarily attributable to advancing age rather than pubertal processes.

Table 1: Robustness check results for the four primary hypothesis tests.

Robustness check	H1 (<i>p</i>)	H2 (<i>p</i>)	H4 (<i>p</i>)
Primary analysis	.307	.310	.988
Spearman rank correlation (ρ)	—	—	—
OLS with robust standard errors	.301	—	—
Time-varying PDS covariate	—	.333	—
Outlier exclusion (± 3 <i>SD</i>)	.348	.383	—
No IQ covariates	.306	—	—
Within-sex model (male)	—	.610	—
Within-sex model (female)	—	.476	—

Note. H3 contrast tests (inhibition vs. working memory) remained non-significant across all robustness specifications (all *ps* > .24). Dashes indicate the check was not applicable to that hypothesis.

Our findings stand in partial tension with the prevailing theoretical model of nonlinear EF development. The canonical decelerating trajectory, first established in cross-sectional studies (??) and meta-analytic syntheses (?), was not replicated in these longitudinal data. This divergence may reflect the cross-sectional versus longitudinal distinction: cross-sectional comparisons conflate age effects with cohort differences, whereas within-person longitudinal models provide a more direct estimate of developmental change (?). It is possible that the quadratic pattern observed in prior cross-sectional work partly reflects cohort effects or differential attrition rather than genuine developmental deceleration. Alternatively, the deceleration may emerge later in development (after age 16) than our data window captured, or may require more than four measurement waves to detect. The absence of pubertal effects on cool EF replicates a recurring pattern in the literature: meta-analytic estimates of the puberty–cool EF link are small and inconsistent (?), and our results suggest that when age is adequately controlled, pubertal stage contributes negligible unique variance to cool EF trajectories. Consistent with the hot-cool distinction (?), pubertal hormones may influence affectively charged decision-making and reward processing more than the abstract cognitive control measured by the NIH Toolbox. The null sex moderation result, while counter to our prediction, aligns with studies showing that sex differences in pubertal timing do not translate into differential cool EF development when age is properly modeled (?).

Several limitations warrant consideration. First, our data are synthetic and were generated to approximate the ABCD Study’s statistical properties; they do not represent real participant data. These results require replication with authenticated ABCD Study data under a valid NDA Data Use Certification before any substantive conclusions can be drawn. Second, the NIH Toolbox measures only cool EF—abstract cognitive control in emotionally neutral contexts. Pubertal effects may be larger for hot EF paradigms (e.g., affective Go/No-Go, delay discounting with real rewards), and our null results should not be interpreted as evidence against puberty–EF links more broadly. Third, the PDS is a self-report measure with only moderate convergence with physician Tanner staging ($r = .61-.80$), and measurement error in pubertal assessment may have attenuated true effects. Fourth, our sample, while diverse by U.S. standards, is a WEIRD cohort that may not generalize to non-Western populations. Fifth, the age range (9–16 years) may miss both the onset of pubertal effects (which may emerge earlier) and the deceleration phase (which may occur after age 16). Finally, the use of random-intercept models (rather than fully random-slope models) may have reduced sensitivity to individual differences in change, though random-slope models showed boundary issues in this dataset. In accordance with open-science practices, the analysis code and synthetic data are publicly available, and the analysis plan was pre-registered prior to hypothesis testing. We encourage reanalysis with real ABCD data and extension to hot EF measures before theoretical conclusions are drawn.